

CSCI 160 Homework 2

1. Demonstrate the validity of the following identity by means of a truth table:

$$x' + yz = (x' + y)(x' + z).$$

2. Simplify each of the following Boolean expressions to a minimum number of literals:

(a) $A'B' + A'B + AB$

(b) $(x + y)(x + y')$

(c) $wx'y + wx'y' + wx$

3. Determine whether the following Boolean equation is true or false. Justify your answer using the postulates and theorems of Boolean algebra (do *not* use a truth table):

$$xy + y'z + xz = xy + y'z.$$

4. Find the complement of the function $F = w'x + yz'$. Then show that $F \cdot F' = 0$ and $F + F' = 1$.
5. Obtain the truth table of the function $F(x, y, z) = x'y + xz$, and express the function both as a sum of minterms and as a product of maxterms. Give each canonical form in expanded notation as well as in the shorthand $\sum m(\cdot)$ and $\prod M(\cdot)$ notation.
6. Convert each of the following Boolean expressions into both sum-of-products form and product-of-sums form:

(a) $(A' + B)(B + C)$

(b) $w + x(y + z')$

7. We can perform logical operations on strings of bits by treating each pair of corresponding bits separately (a bitwise operation). Given the two eight-bit strings

$$A = 11010110 \quad \text{and} \quad B = 10011010,$$

evaluate the eight-bit result of the bitwise XOR ($A \oplus B$) and the bitwise NOR of A and B .

8. Without simplifying the expression, draw the logic diagram that implements the Boolean function

$$F = AB' + A'B + BC$$

using AND, OR, and inverter (NOT) gates. Label your inputs and mark the output as F .